BAIO LEARNING SYSTEM • 3-DAY TARGET REVISION

Worksheet #3 of 3

Day 3 Worksheet: Leaf Anatomy & Comprehensive 4-Mark Board Exam Test

Dicot vs Monocot Leaf | Bulliform Cells | Full Chapter 4-Mark Board Pattern Mastery

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Student Name: .... **Date:** .... **Time Allowed:** 30 Mins **Max Marks:** 20 **Marks Obtained:** ....

Day 3 Focus Review: • Dicot (Dorsiventral) Leaf: Hypostomatic, Mesophyll differentiated into Palisade & Spongy parenchyma, reticulate venation. • Monocot (Isobilateral) Leaf: Amphistomatic, Undifferentiated mesophyll, *Bulliform cells* on adaxial epidermis (induce rolling to limit transpiration). • Master Chapter Board Blueprint: Total 4 Marks (`3M + 1M` or `2M + 2M`).

SECTION A: CONCEPT RECALL & ONE-LINERS (1 MARK EACH)

1. Large, empty, colourless cells present on the adaxial epidermis of grass leaves are called: 1M

(a) Guard cells (b) Bulliform cells (c) Subsidiary cells (d) Lenticels

2. In a dorsiventral leaf, the palisade parenchyma is located towards the: 1M

(a) Abaxial epidermis (b) Adaxial epidermis (c) Midrib vein only (d) Stomatal cavity

3. Name the tissue that performs photosynthesis in leaves. 1M

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4. Which type of venation is characteristically found in monocotyledonous leaves? 1M

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SECTION B: SHORT ANALYTICAL QUESTIONS (2 MARKS EACH)

5. Explain the mechanical action of Bulliform cells during water deficit / drought in grasses. 2M

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6. What are Casparian strips? State their chemical composition and exact anatomical location. 2M

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7. Differentiate between Palisade parenchyma and Spongy parenchyma. 2M

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SECTION C: 4-MARK BOARD EXAM MOCK TEST (TARGET 4/4)

8. Enumerate three distinct anatomical differences between a Dorsiventral (Dicot) leaf and an Isobilateral (Monocot) leaf. 3M

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9. [Annual Board 4-Mark Challenge]: 4M

(a) Draw a labeled diagram of the Stomatal apparatus of a grass leaf. (2M)

(b) Why are vascular bundles in monocot stems termed "closed"? (1M)

(c) Name the layer referred to as 'starch sheath' in dicot stem. (1M)

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